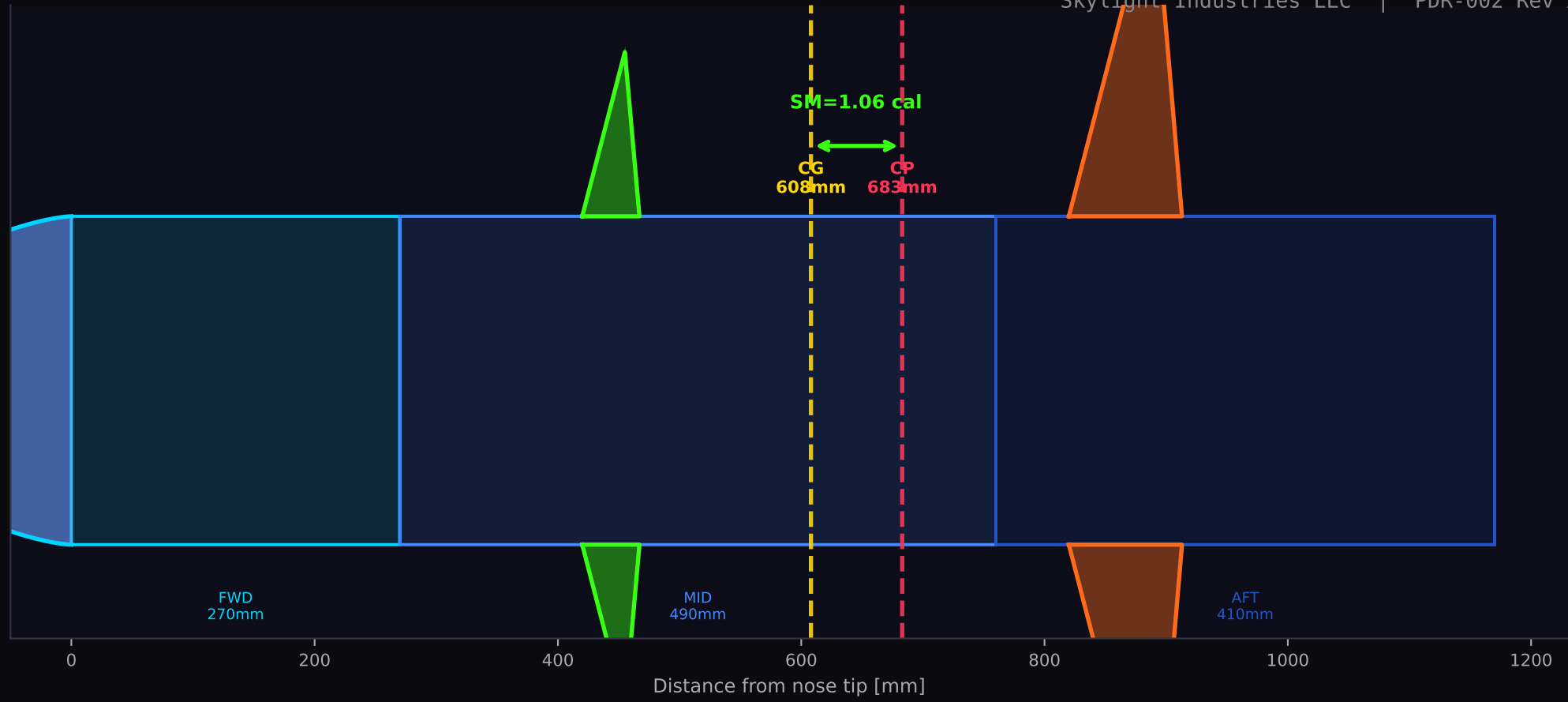
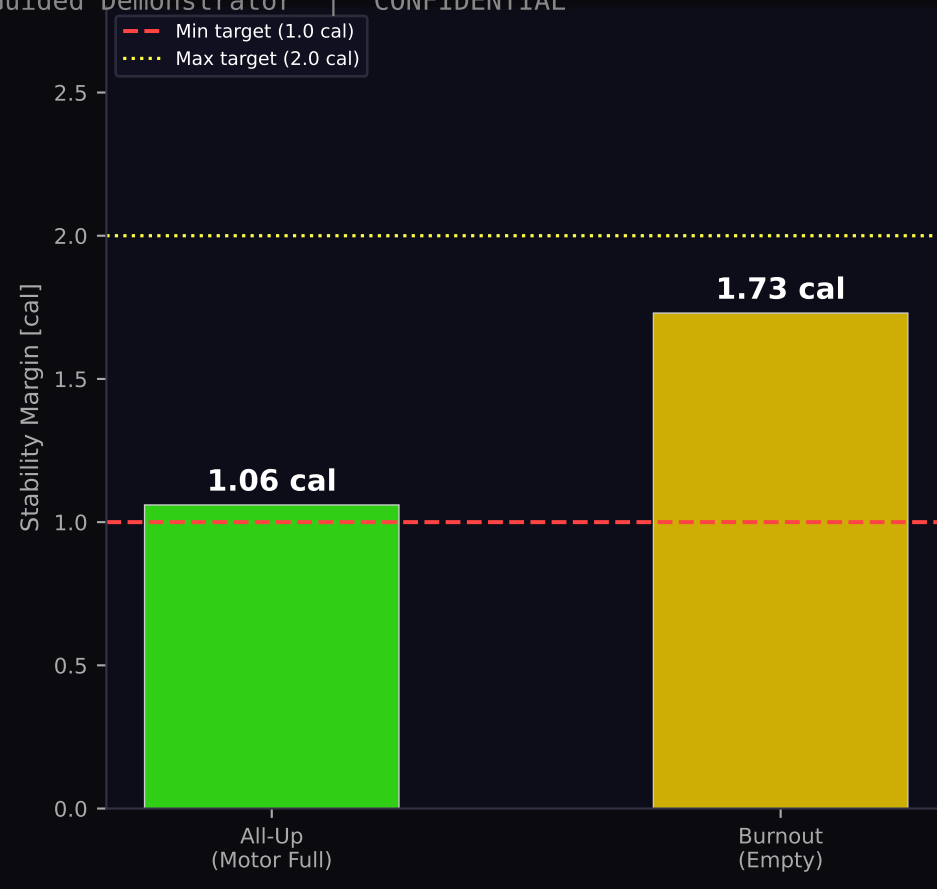
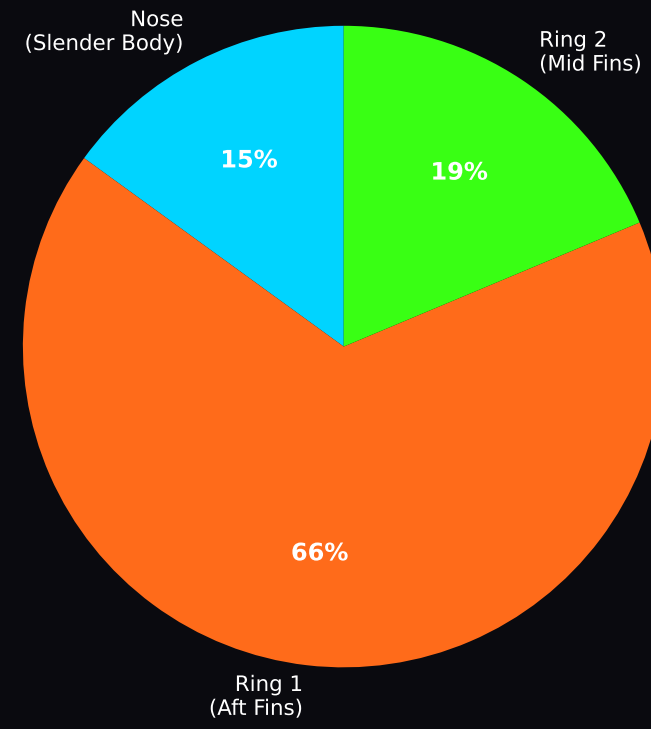


XRIM-117 WYVERN-E - ENGINEERING ANALYSIS REPORT

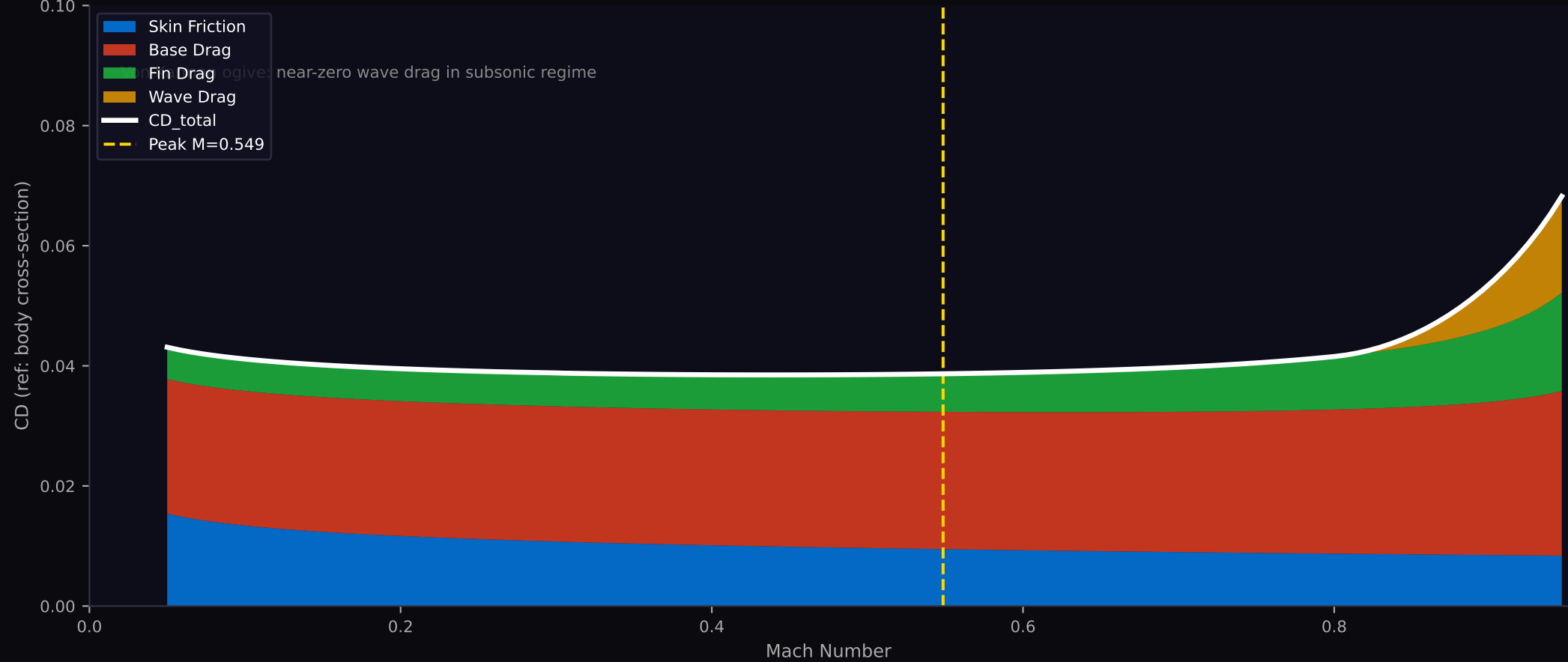
VEHICLE PROFILE & STABILITY



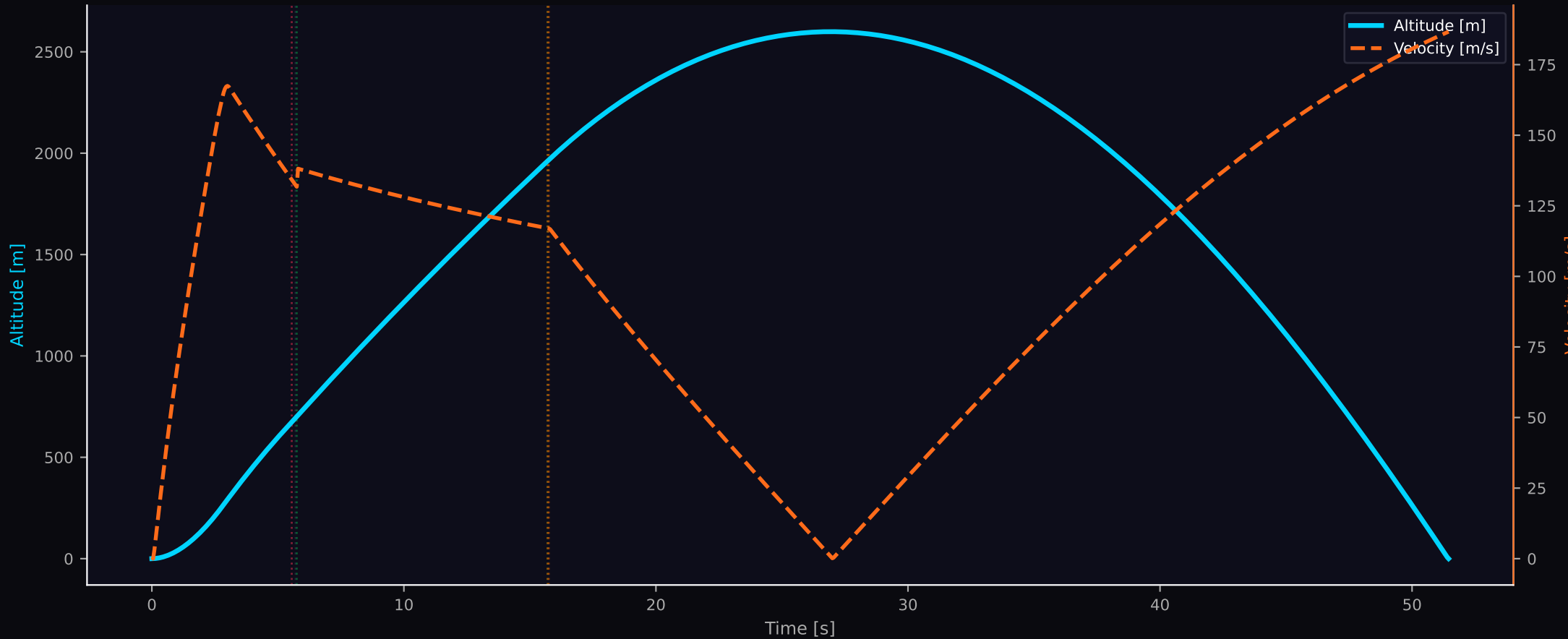
STABILITY MARGIN

CN α BREAKDOWN
Total = 13.33/rad

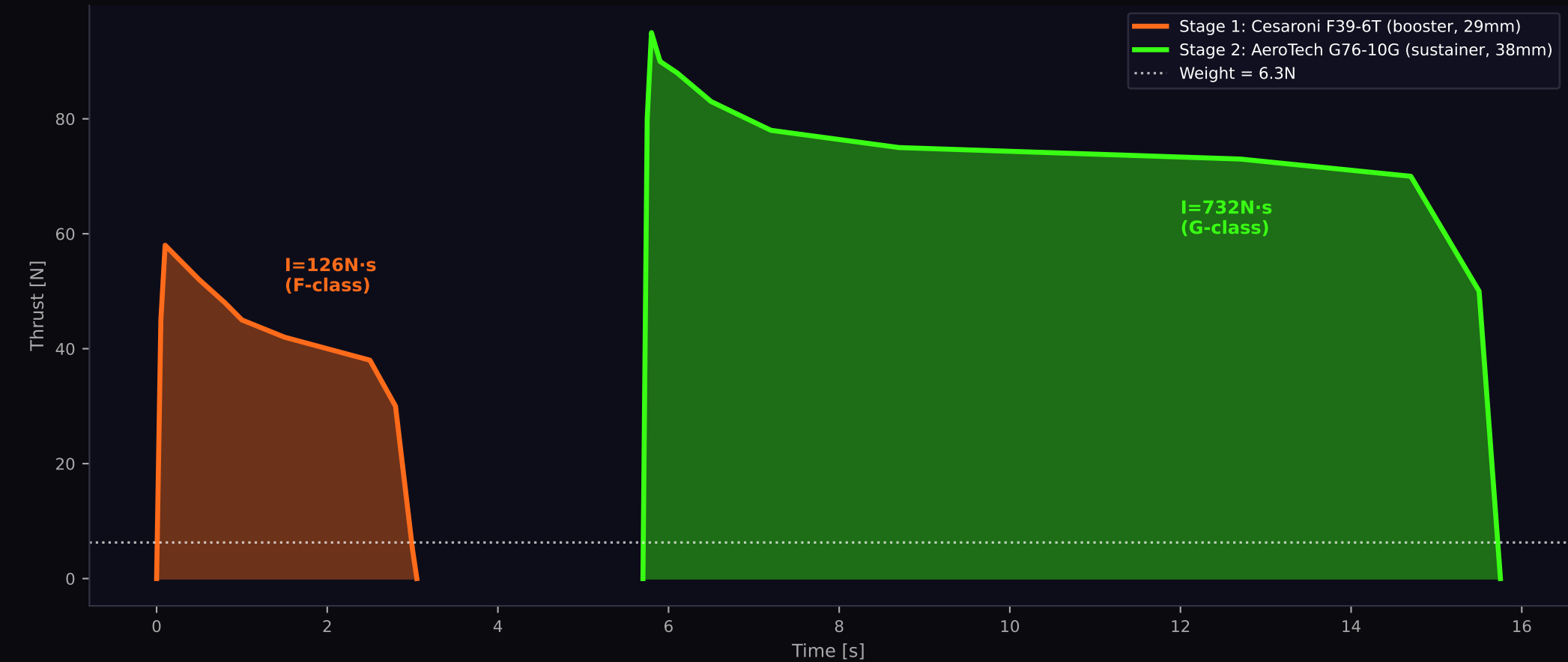
DRAG COEFFICIENT vs MACH — COMPONENT BREAKDOWN



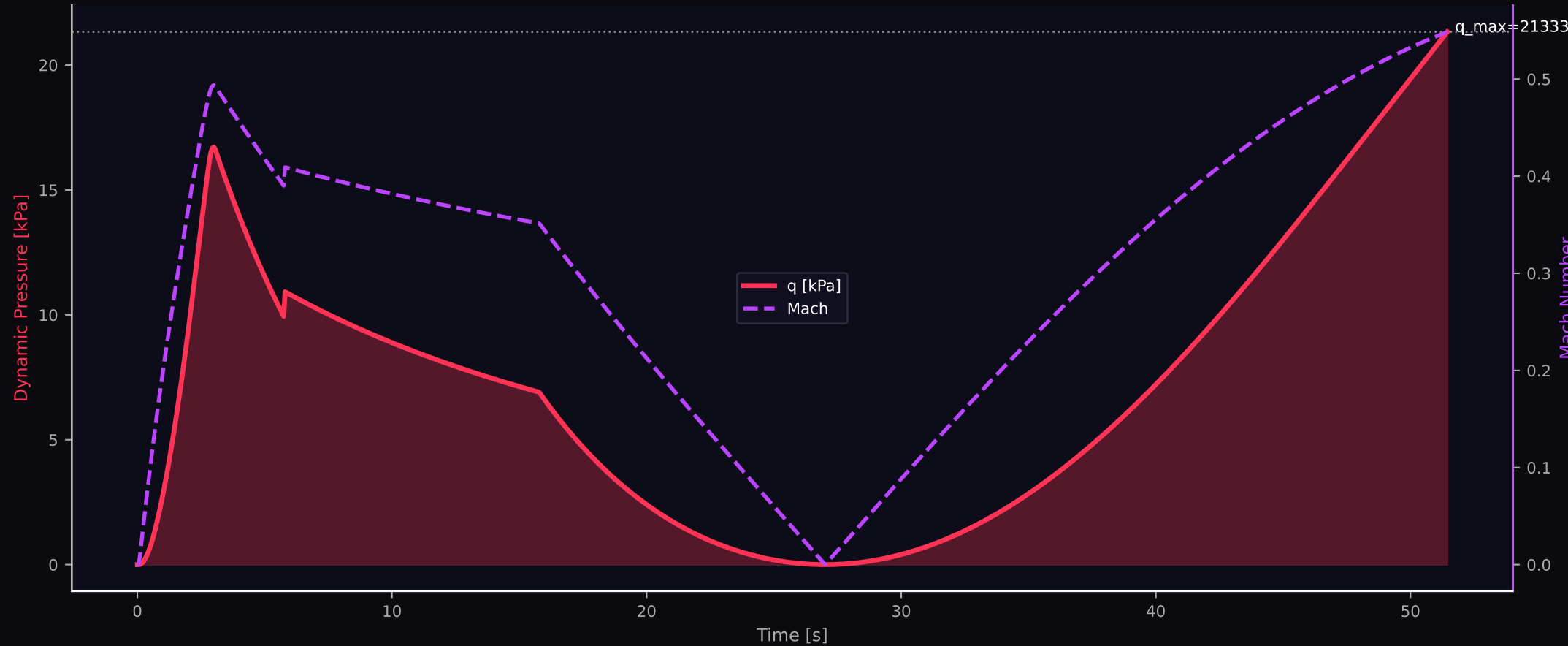
ALTITUDE & VELOCITY vs TIME — TRAJECTORY SIMULATION



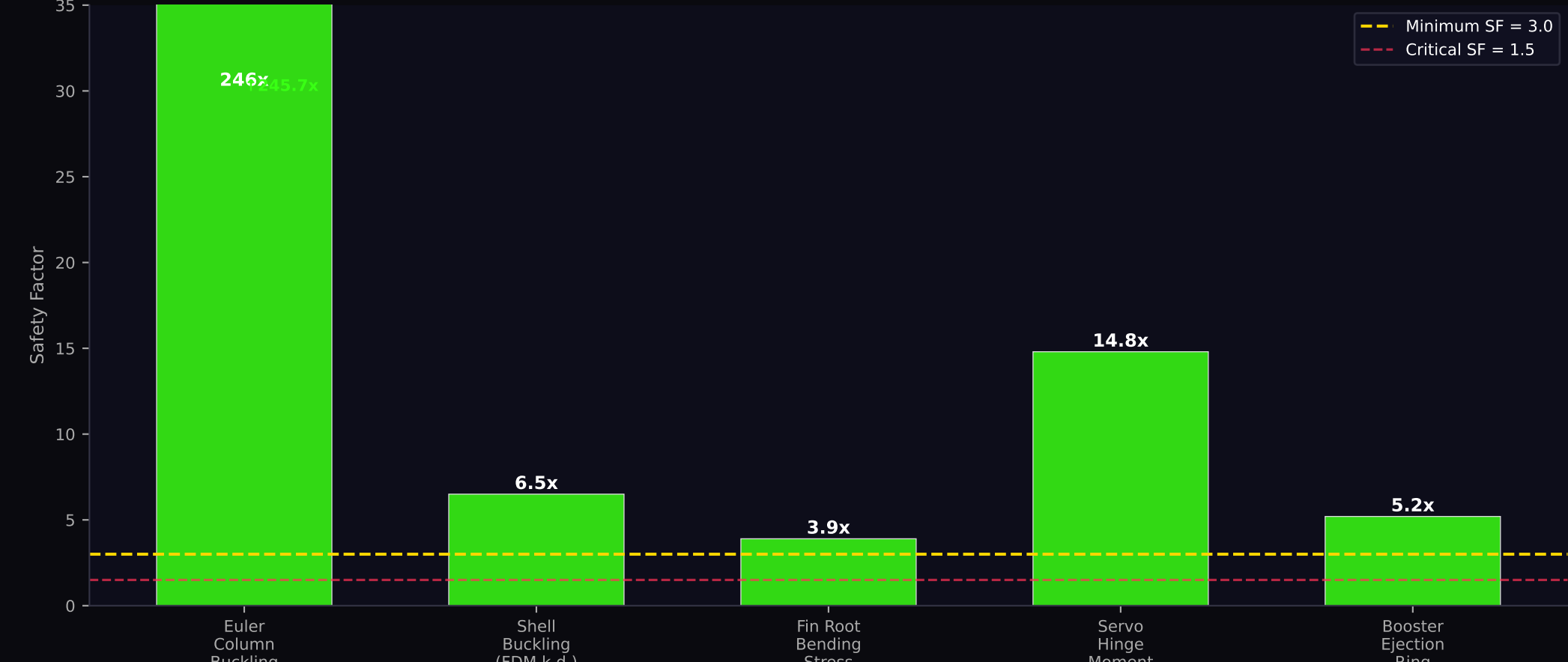
MOTOR THRUST CURVES — TWO-STAGE PROPULSION



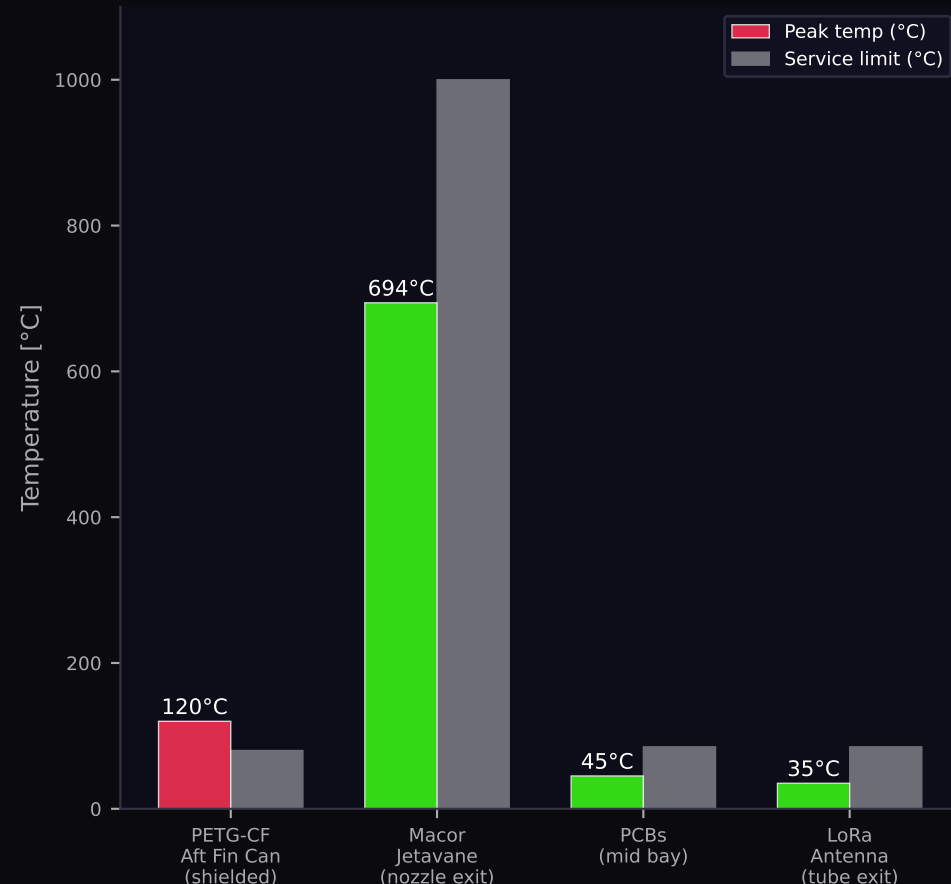
DYNAMIC PRESSURE & MACH vs TIME



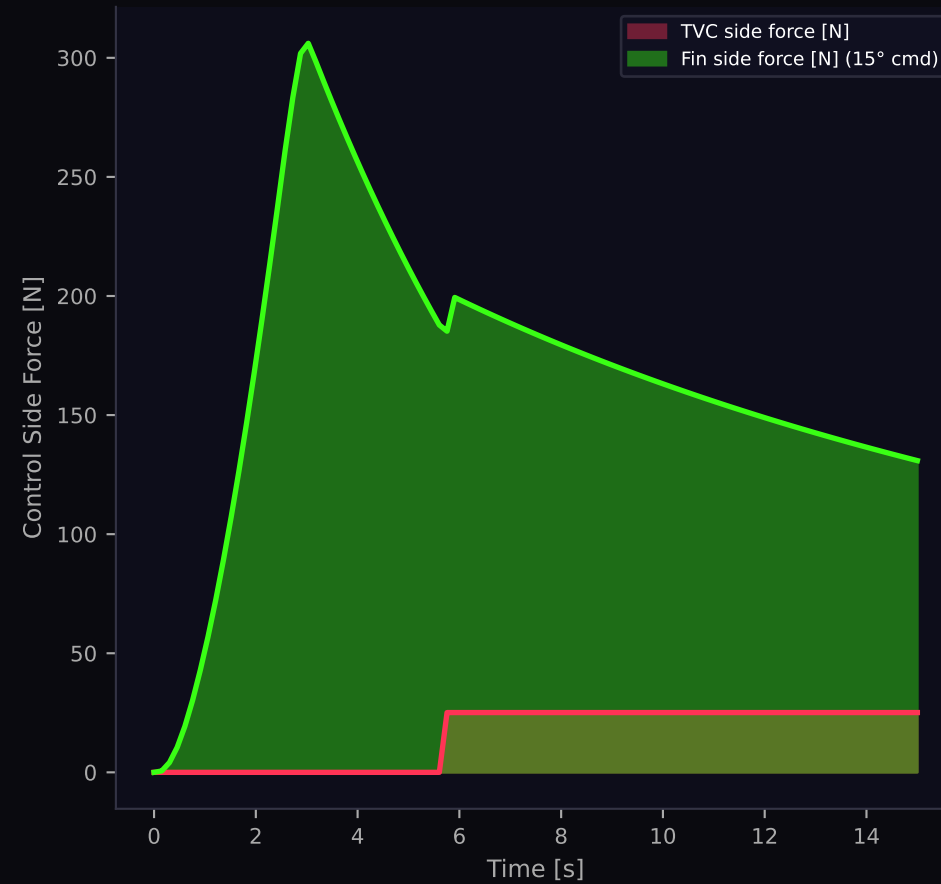
FEA SAFETY FACTORS — STRUCTURAL ANALYSIS



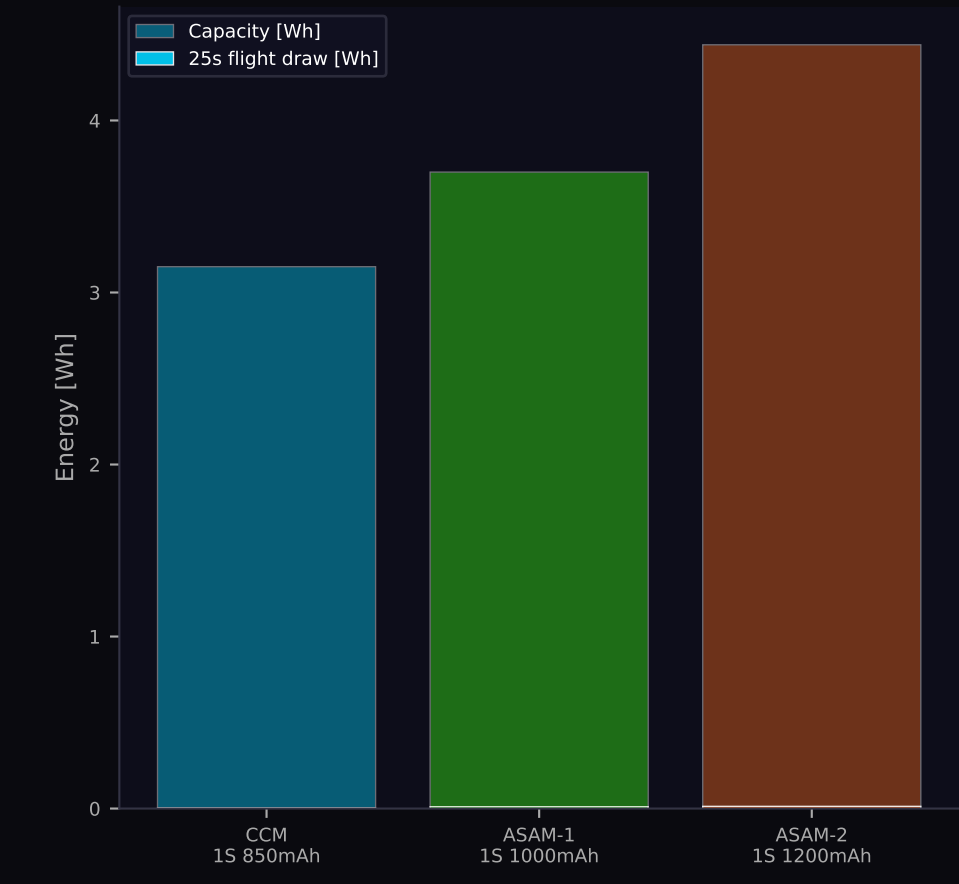
THERMAL MARGINS



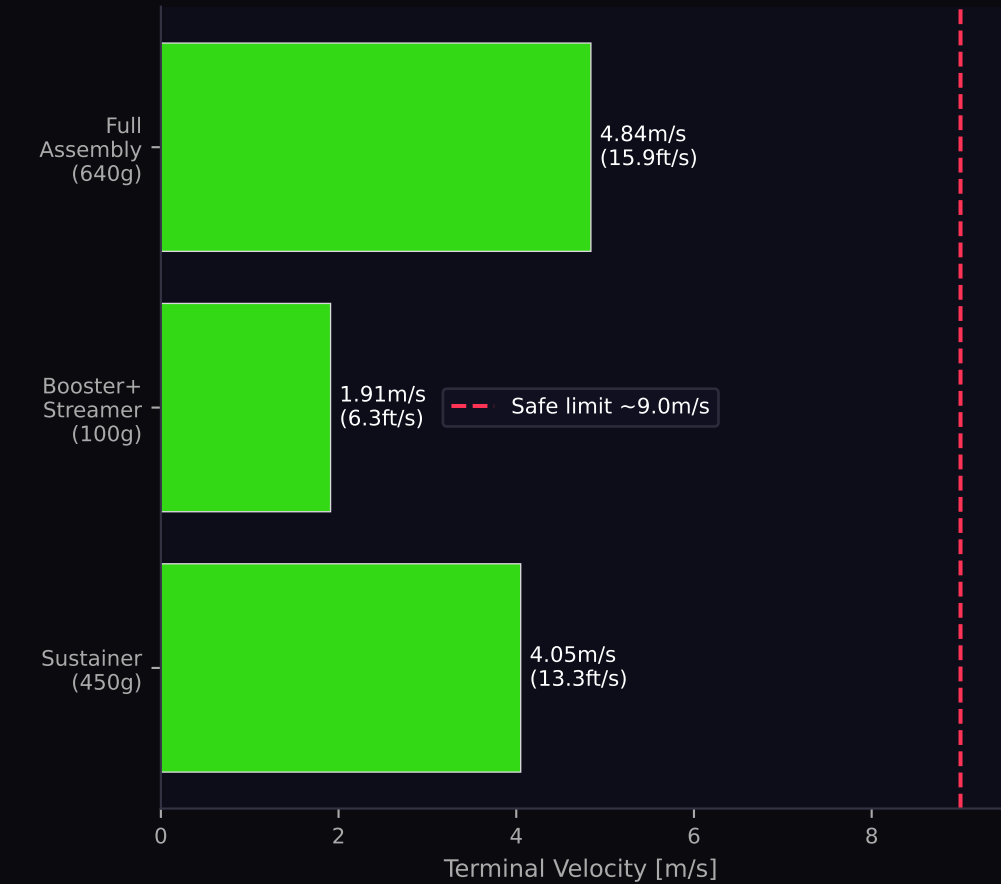
CONTROL AUTHORITY — TVC vs FINS



POWER BUDGET



RECOVERY — TERMINAL VELOCITIES



ENGINEERING SUMMARY — KEY PARAMETERS

STABILITY	SM=1.06 cal (full) / 1.73 cal (burnout)	✓ PASS
AERODYNAMICS	CD=0.0388 @ M=0.3 q_max=21333Pa	✓ PASS
PROPULSION	I_total=858N·s T/W_Liftoff=9.2x	✓ PASS
STRUCTURE	Shell buckling SF=6.5x Fin root SF=3.9x	✓ PASS
TVC THERMAL	Macor @ 694°C exit gas (<1000°C limit)	✓ PASS
FIN THERMAL	Aft fin can ~120°C (foil shield reqd)	⚠ SHIELD REQ
SERVOS	Hinge moment 38mN·m vs 441mN·m cap 14.8x margin	✓ PASS
CONTROL BW	25Hz control loop 500Hz ASAM servo ICM-42688-P 32kHz	✓ PASS
RECOVERY	24" chute v_term=4.84m/s all-up KE=7.5J touchdown	✓ PASS
DATALINK	LoRa 915MHz link margin=63dB range=2km LOS	✓ PASS
POWER	3 independent LiPo flight=25s standby>2hr	✓ PASS
TRAJECTORY	Peak M=0.494 v_max=167m/s z_max=150m (guided)	✓ PASS